

HUM 4201: PRINCIPLES OF ECONOMICS

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Chapter-01: Measuring a Nation's Income

ECONOMICS

1 Economics is the science of wealth
(Adam Smith, 1776) [Father of Economics]

2 Economics is the science which studies human behavior as a relationship between ends and scarce means which have alternative uses
(Lionel Robbins, 1932)



Means: Goods and services that are available

Ends: Needs and requirements that are satisfied through the means

INTRODUCTION

Real-World Motivation

- ❑ When you finish school and start looking for a full-time job, your experience depends heavily on **economic conditions**.
- ❑ In some years:
 - Firms expand production
 - Employment rises
 - Jobs are easy to find
- ❑ In other years:
 - Firms cut production
 - Employment declines
 - Jobs are difficult to obtain

👉 Therefore:

Economic conditions directly shape individual outcomes in the labor market.

INTRODUCTION

Expansion vs Contraction

- **Economic Expansion:**
 - ✓ Rising production
 - ✓ Increasing employment
 - ✓ Higher job availability
 - **Economic Contraction:**
 - ✓ Falling production
 - ✓ Declining employment
 - ✓ Longer job search time
-  **Key insight:**
- Individuals prefer to enter the labor market during **expansionary periods** rather than **contractionary periods**.

Why Study the Overall Economy?

- The **health of the overall economy profoundly affects all of us.**
 - Economic changes are:
 - Widely reported in newspapers
 - Covered in online platforms
 - Broadcast on television
-  **Conclusion:**
- Understanding macroeconomic indicators is essential for interpreting real-world events.

INTRODUCTION

Macroeconomic Indicators in the News

Commonly reported statistics include:

- **Gross Domestic Product (GDP)** → total income of the economy
- **Inflation / Deflation** → changes in average prices
- **Unemployment rate** → percentage of labor force without jobs
- **Retail sales** → total spending at stores
- **Trade deficit** → imbalance in international trade

 These are all **macroeconomic statistics**, reflecting the entire economy rather than individual markets.

INTRODUCTION

Microeconomics vs Macroeconomics

Microeconomics

- Study of:
 - Individual households
 - Firms
 - Market interactions

Microeconomics the study of how households and firms make decisions and how they interact in markets

Macroeconomics

- Study of:
 - Economy-wide phenomena
 - Inflation
 - Unemployment
 - Economic growth

Macroeconomics the study of economy-wide phenomena, including inflation, unemployment, and economic growth

INTRODUCTION

Core Questions of Macroeconomics

Macroeconomists seek to answer:

- Why is **average income higher in some countries** than others?
- Why do **prices sometimes rise rapidly** (inflation)?
- Why do **production and employment fluctuate**?
- What policies promote:
 - ✓ Economic growth
 - ✓ Low inflation
 - ✓ Stable employment

👉 These are **economy-wide questions**, not individual-level issues.

Link Between Micro and Macro

❖ The economy consists of:

- Many households
- Many firms
- Many markets

❖ Therefore:

- **Microeconomic tools (e.g., supply & demand)** are fundamental to macroeconomics

📌 However:

- Studying the entire economy introduces **new analytical challenges**.

INTRODUCTION

Purpose of This Chapter

- This chapter introduces **key macroeconomic data** used by:
 - ✓ Economists
 - ✓ Policymakers
- Focus:
 - ✓ Measuring economic performance
 - ✓ Understanding aggregate indicators

Central Focus – GDP

- **Gross Domestic Product (GDP):**
 - ✓ Measures **total income of a nation**
 - ✓ Considered the **single best measure of economic well-being**
- 📌 **Why GDP matters:**
 - ✓ Reflects overall economic activity
 - ✓ Used to evaluate economic performance
 - ✓ Closely monitored worldwide

THE ECONOMY'S INCOME AND EXPENDITURE

Intuition – Individual vs Economy

❑ To judge a **person's economic well-being**:

- We look at **income**
- Higher income → higher standard of living

❑ Examples of higher living standards:

- Better housing
- Better healthcare
- Luxury goods
- Vacations

👉 **Same logic applies to the entire economy**

From Individual to National Level

❑ For a nation:

- Economic performance is judged by **total income of everyone**

📌 This is measured by:
Gross Domestic Product (GDP)

THE ECONOMY'S INCOME AND EXPENDITURE

Dual Nature of GDP

GDP measures TWO things:

1. **Total income of everyone in the economy**
2. **Total expenditure on goods and services**

👉 Key identity:

Income = Expenditure

Implication for GDP

- GDP increases by the same amount whether measured by:
- Income approach
 - Expenditure approach

📌 Therefore: **GDP = Total Income = Total Expenditure**

Why Income = Expenditure?

- Every economic transaction has:
 - A **buyer**
 - A **seller**
- Therefore:
 - Buyer's spending = Seller's income

📌 Example:

- Karen pays Doug \$100 to mow lawn
 - Karen → spends \$100
 - Doug → earns \$100
- 👉 Contribution to GDP:
 - Income = \$100
 - Expenditure = \$100

THE ECONOMY'S INCOME AND EXPENDITURE

Circular Flow Framework

The Circular-Flow Diagram explains this equality

- ❑ Shows interaction between:
 - ✓ **Households**
 - ✓ **Firms**
- ❑ Two types of markets:
 - ✓ Goods & Services Market
 - ✓ Factors of Production Market

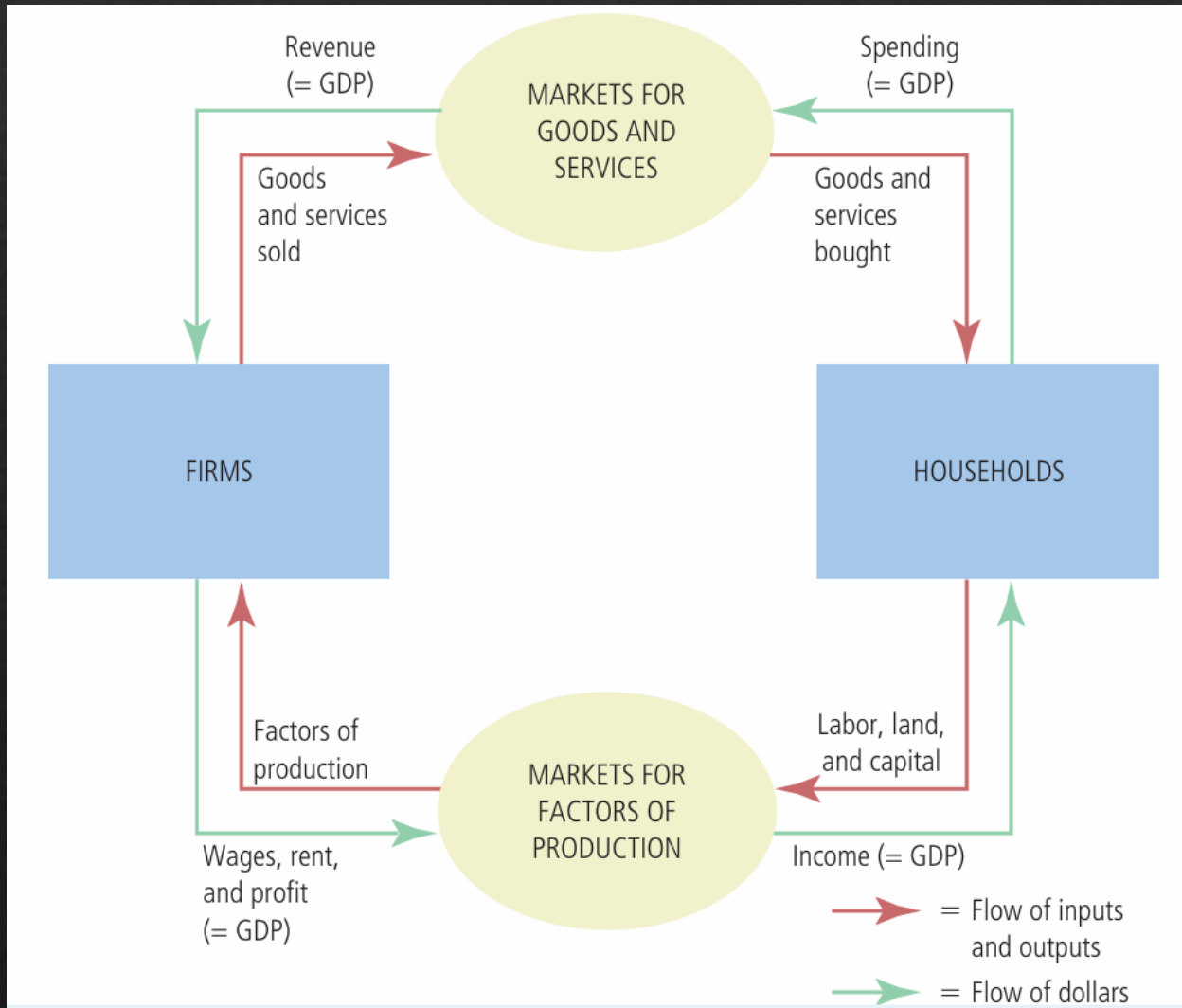
Circular Flow – Factor Market

- Firms:
 - Pay wages, rent, profit
- Households:
 - Supply labor, land, capital
- 👉 Flow:
 - **Firms → Households (Income)**

Circular Flow – Goods Market

- ❖ Households:
 - Buy goods and services
 - Spend money
- ❖ Firms:
 - Sell goods and services
 - Receive revenue
- 👉 Flow:
 - **Households → Firms (Spending)**

THE ECONOMY'S INCOME AND EXPENDITURE



Visual Summary

Key flows in the economy:

- **Goods flow:** Firms → Households
- **Factor services:** Households → Firms
- **Money flow (spending):** Households → Firms
- **Money flow (income):** Firms → Households

👉 GDP measures **these monetary flows**

THE ECONOMY'S INCOME AND EXPENDITURE

Simplifying Assumptions (Basic Model)

In the simple circular-flow model:

- Households:
 - ✓ Spend all income
- Firms:
 - ✓ Produce all goods



No:

- Government
- Savings
- Investment
- Foreign trade

Real-World Economy (More Complex)

In reality:

- Households:
 - ✓ Pay taxes
 - ✓ Save part of income
- Firms:
 - ✓ Invest
- Government:
 - ✓ Purchases goods/services
- Foreign sector:
 - ✓ Imports & exports



BUT:

❖ **The core identity still holds:**



Income = Expenditure

THE MEASUREMENT OF GDP

Formal Definition of GDP

Gross Domestic Product (GDP):

“The market value of all final goods and services produced within a country in a given period of time.”

 Each word in this definition is **economically significant**

We analyze GDP through its components:

1. **Market value**
2. **All**
3. **Final**
4. **Goods and services**
5. **Produced**
6. **Within a country**
7. **Given period of time**

THE MEASUREMENT OF GDP

PART A: “GDP is the Market Value ...”

Market Value Concept

- GDP aggregates **different goods** (apples, cars, healthcare, etc.)
- Uses **market prices** as weights

👉 Why?

- Prices reflect **willingness to pay**
- Allow **comparison across heterogeneous goods**

📌 Example:

- If apple price = 2 × orange price
→ Apple contributes twice as much to GDP

PART B: “... of All ...”

Comprehensiveness

- GDP includes:
 - ✓ All goods and services **produced and sold legally in markets**
- Examples:
 - ✓ Food, clothing, movies
 - ✓ Haircuts, healthcare

THE MEASUREMENT OF GDP

PART B: "... of All ..."

Imputed Values

- Some items are **included even without market transactions**
- **Example:**
 - ☑ Owner-occupied housing
 - ✓ No rent paid
 - ✓ Government imputes rental value

Paradox Example

- If Karen pays Doug to mow lawn → included in GDP
 - If Karen marries Doug → excluded
- 👉 Same activity, different treatment

📌 Insight:

- GDP depends on **market transactions**, not actual output alone

What GDP Excludes

☐ GDP excludes:

- ✓ Illegal goods (e.g., drugs)
- ✓ Household production:
 - Home-grown vegetables
 - Domestic services



Key implication:

- GDP may **underestimate actual economic activity**

THE MEASUREMENT OF GDP

PART C: "... Final ..."

Final vs Intermediate Goods

- ❑ **Intermediate goods:**

- ✓ Used to produce other goods

- ❑ **Final goods:**

- ✓ Ready for final use

 GDP includes: **ONLY final goods**

Inventory Exception

- If intermediate goods are **unsold**:
 - ✓ Treated as **inventory investment**

 Implication:

- ✓ Added to GDP when produced
- ✓ Subtracted when sold later

Avoiding Double Counting

- Example:

- ✓ Paper → Greeting card

- If both counted:

- ✓ Value counted twice ❌

 Solution:

- Count only **final good (card)**

THE MEASUREMENT OF GDP

PART D: "... Goods and Services ..."

GDP includes:

- ❑ **Tangible goods:**

- ✓ Food, cars, clothing

- ❑ **Intangible services:**

- ✓ Healthcare, education, concerts

 Both contribute equally to GDP

PART E: "... Produced ..."

- GDP includes:

- Goods/services produced **in current period**

- Excludes:

- Used goods (second-hand sales)

 Example:

- New car → included
 - Used car resale → excluded

THE MEASUREMENT OF GDP

PART F: "... Within a Country ..."

- GDP depends on **location of production**, not nationality
- Examples:
 - Foreign worker in Bangladesh → counted in BD GDP
 - Bangladesh firm in Nepal → NOT counted in Bangladesh GDP
- 📌 Key distinction:
 - ✓ GDP ≠ GNP

PART G: "... In a Given Period of Time"

- GDP is a **flow variable**
- Measured over:
 - ✓ Year
 - ✓ Quarter

Two Methods of Measuring GDP

- GDP can be calculated by:
 1. **Expenditure approach**
 2. **Income approach**
- 👉 Both give (almost) same result

OTHER MEASURES OF INCOME

- GDP is the most widely used indicator
- But economists also use **alternative income measures**
- Purpose:
 - ✓ To get a **more complete picture of the economy**

List of Income Measures (Largest → Smallest)

1. Gross National Product (GNP)
2. Net National Product (NNP)
3. National Income (NI)
4. Personal Income (PI)
5. Disposable Personal Income (DPI)

Gross National Product (GNP)

- **GNP**: Total income earned by a nation's **permanent residents (nationals)**
- **GDP vs GNP**
 - **GDP**: Based on **location of production**
 - **GNP**: Based on **ownership (citizenship)**
- Differences:
 - Includes: Income earned by citizens **abroad**
 - Excludes: Income earned by foreigners **domestically**
- **Example**:
 - Canadian working in USA:
 - ✓ Included in **US GDP**
 - ✓ Included in **Canada's GNP**

OTHER MEASURES OF INCOME

Net National Product (NNP)

❖ **NNP:**

$$NNP = GNP - Depreciation$$

❖ Depreciation:

- Wear and tear of capital
- Also called **consumption of fixed capital**

❖ NNP Shows:

- **Net output after maintaining capital stock**

❖ Insight: Reflects **sustainable income level**

National Income (NI)

➤ Total income earned by residents from production

➤ **NI vs NNP**

- Nearly identical
- Difference arises due to:
 - ✓ **Statistical discrepancy**
- Reason: Imperfect data collection

OTHER MEASURES OF INCOME

Personal Income (PI)

- ❑ Income received by:
 - Households
 - Noncorporate businesses
- ❑ **Adjustments in PI:** Compared to National Income:
 - ❖ **Subtract:**
 - Retained earnings (corporate profits not distributed)
 - Corporate taxes
 - Social insurance contributions
 - ❖ **Add:**
 - Government transfers
 - Interest on government debt
- ❑ PI Reflects: **Actual income received by households**
- ❑ More relevant for: Consumption decisions

Disposable Personal Income (DPI)

- ❑ $DPI = PI - \text{Personal Taxes}$
- ❑ DPI means income available for:
 - Consumption
 - Saving
- ❑ Most relevant for: Household welfare

OTHER MEASURES OF INCOME

Summary Table

Measure	Adjustment	Meaning
GDP	-----	Domestic production
GNP	+ Net foreign income	National income
NNP	– Depreciation	Net output
NI	Adjusted NNP	Income earned
PI	Adjusted NI	Income received
DPI	– Taxes	Income available

THE COMPONENTS OF GDP (IN EXPENDITURE METHOD)

○ Spending in an economy takes many forms:

- ✓ At any moment, your family may be having lunch at PizzaBurg (**Household consumption**);
- ✓ Walton may be building a car factory (**Firms/Farms investment**);
- ✓ The Bangladesh Navy may be procuring a submarine (**Govt. expenditure**); and
- ✓ Biman Bangladesh Airlines may be buying an airplane from Boeing (**Foreign trade**).
- ✓ GDP includes all of these various forms of spending on domestically produced goods and services.

GDP Components

$$Y = C + I + G + NX$$

Where:

- Y = Gross Domestic Product
- C = Consumption
- I = Investment
- G = Government Purchases
- NX = Net Exports

THE COMPONENTS OF GDP (IN EXPENDITURE METHOD)

Consumption (C)

- Spending by households on:
 - ✓ Goods
 - ✓ Services
- Exception:
 - ✓ Purchase of **new housing** is NOT consumption

Important Clarification

- Education spending → counted as **consumption**
- Even though:
 - ✓ It may increase future productivity
- Conceptual note:
 - ✓ Classification is based on **current use**, not future benefit

Types of Consumption Goods

1. Durable goods

- Long-lasting
- Example: cars, appliances

2. Nondurable goods

- Short-lived
- Example: food, clothing

3. Services

- Intangible
- Example: healthcare, education

THE COMPONENTS OF GDP (IN EXPENDITURE METHOD)

Investment (I)

- ❖ Purchase of goods (called capital goods) that will be **used in the future to produce more goods and services**.
- ❖ Investment is the sum of purchases of business capital, residential capital, and inventories.
- ❖ **Business capital includes:**
 - Business structures (such as a factory or office building),
 - Equipment (such as a worker's computer), and
 - Intellectual property products (such as the software that runs the computer).
 - Residential capital includes the landlord's apartment building and a homeowner's personal residence.
 - By convention, the **purchase of a new house** is the one type of household spending categorized as investment rather than consumption.

THE COMPONENTS OF GDP (IN EXPENDITURE METHOD)

Investment (I)

Inventory Mechanism

- ❑ If a firm produces but does not sell (i.e., computer):
 - Counted as **investment** ((treated as: purchase of goods (called capital goods))
- ❑ Later sale:
 - Inventory decreases
 - Investment becomes negative
- ❑ Ensures: GDP reflects **production, not just sales**

Key Distinction

- ❖ **Investment in GDP $\neq$ Financial Investment**
 - ✓ GDP investment includes:
 - Physical capital
 - ✓ Does NOT include:
 - Stocks
 - Bonds
 - Mutual funds
 - ✓ Reason: These are **financial transactions**, not production

THE COMPONENTS OF GDP (IN EXPENDITURE METHOD)

Government Purchases (G)

- ❖ Spending by government on Goods & Services
- ❖ **What is Included**
 - ☑ Salaries of:
 - Teachers
 - Military personnel
 - ☑ Public infrastructure:
 - Roads
 - Bridges

Transfer Payments (Excluded)

- When the government pays a Social Security benefit to a person who is elderly or an unemployment insurance benefit to a worker who was recently laid off, the story is very different:
- These are called **transfer payments** because they are not made in exchange for a currently produced good or service.
- Transfer payments alter household income, but they do not reflect the economy's production.
- Because GDP is intended to measure income from, and expenditure on, the production of goods and services, transfer payments are not counted as part of government purchases.

THE COMPONENTS OF GDP (IN EXPENDITURE METHOD)

Net Exports (NX)

- Net Exports equal the foreign purchases of domestically produced goods (exports) minus the domestic purchases of foreign goods (imports).

$$NX = \text{Exports} - \text{Imports}$$

- Exports: Increase GDP
- Imports: Reduce GDP
- Reason: Imports are included in C, I, G → must subtract

❖ But is this car produced domestically?

- ✓ No → It is produced in Sweden (foreign country)
- ✓ So, it should **NOT** be counted in domestic GDP

Step-by-Step Simple Example

- ❖ **Situation:** You buy a car for \$40,000 from Volvo (Sweden).
- ❖ **What happens to Consumption (C)?**
 - ✓ Buying a car = consumption
 - ✓ So: $C = +40,000$

❖ Final Effect on GDP

- ✓ Recall:

$$GDP = C + I + G + NX$$

- ✓ Substitute:

$$GDP = 40,000 + 0 + 0 - 40,000 = 0$$

❖ **Key Insight:** Net effect on GDP: Zero

REAL VS NOMINAL GDP

- ❑ If total spending rises from one year to the next, at least one of two things must be true:
 - (1) the economy is producing a larger output of goods and services, or
 - (2) goods and services are being sold at higher prices.
- ❖ When studying **changes in the economy over time**, We, the economists, want to separate these two effects.
- ❖ In particular, we want a measure of the total quantity of goods and services the economy is producing that is **not affected by changes in the prices (Real GDP)** of those goods and services.
- Real GDP answers a hypothetical question:
 - What would be the value of the goods and services produced this year if we valued these goods and services at the prices that prevailed in some specific year in the past?
- By evaluating current production using prices that are fixed at past levels, **real GDP shows how the economy's overall production of goods and services changes over time.**

REAL VS NOMINAL GDP

Nominal GDP

- ❖ Value of goods and services at **current prices**
- ❖ Formula:

$$\text{Nominal GDP} = \sum (P_t \times Q_t)$$

- ❖ Reflects: Both price changes AND quantity changes

Real GDP

- ❑ Value of goods and services at **constant (base-year) prices**
- ❑ Formula:

$$\text{Real GDP} = \sum (P_{base} \times Q_t)$$

- ❑ Reflects: Only quantity changes

REAL VS NOMINAL GDP

Numerical Example

□ Economy produces:

✓ Hot dogs

✓ Hamburgers

Year	Price (Hot Dog)	Qty	Price (Hamburger)	Qty
2016	\$1	100	\$2	50
2017	\$2	150	\$3	100
2018	\$3	200	\$4	150

□ Base year = 2016

Nominal GDP Calculation

2016:

$$(1 \times 100) + (2 \times 50) = 200$$

2017:

$$(2 \times 150) + (3 \times 100) = 600$$

2018:

$$(3 \times 200) + (4 \times 150) = 1200$$

👉 Nominal GDP rises sharply

Real GDP Calculation

2016:

$$(1 \times 100) + (2 \times 50) = 200$$

2017:

$$(1 \times 150) + (2 \times 100) = 350$$

2018:

$$(1 \times 200) + (2 \times 150) = 500$$

👉 Real GDP increases due to **higher quantities only**

REAL VS NOMINAL GDP

Key Insight

- Nominal GDP:
 - ✓ $200 \rightarrow 600 \rightarrow 1200$
- Real GDP:
 - ✓ $200 \rightarrow 350 \rightarrow 500$

Summary Comparison

Feature	Nominal GDP	Real GDP
Prices used	Current	Base-year
Measures	Value	Quantity
Affected by inflation?	Yes	No
Best for welfare?	No	Yes

- Growth rate = % change in **real GDP**

REAL VS NOMINAL GDP

GDP Deflator

- ❑ From these two (Nominal and Real GDP) statistics, we can compute a third, called the **GDP deflator**, which reflects only the prices of goods and services.
- ❑
$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$
- ❑ Measures: The current level of prices relative to the level of prices in the base year.

GDP Deflator Calculation

2016:

$$(200/200) \times 100 = 100$$

2017:

$$(600/350) \times 100 = 171$$

2018:

$$(1200/500) \times 100 = 240$$

 Base year deflator = 100

Interpretation

- Shows how much prices have changed relative to base year
- Example: 171 → prices increased by **71%** from base year

REAL VS NOMINAL GDP

Inflation Rate

- ❖ Economists use the term inflation to describe a situation in which the economy's overall price level is rising.
- ❖ The **inflation rate is the percentage change** in some measure of the price level from one period to the next.
- ❖ Using the GDP deflator, the inflation rate between two consecutive years is computed as follows:

$$\square \text{Inflation Rate} = \frac{\text{Deflator}_t - \text{Deflator}_{t-1}}{\text{Deflator}_{t-1}} \times 100$$

□ Inflation Calculation

2017:

$$\frac{171 - 100}{100} \times 100 = 71\%$$

2018:

$$\frac{240 - 171}{171} \times 100 \approx 40\%$$

Interpretation

- Inflation reflects:
 - ✓ Rising price level
- GDP deflator:
 - ✓ Captures economy-wide inflation